

PCP SIKAR SMART REVISION SYSTEM

PCP MAHAMUKABLA MASTER DPP NOTEBOOK

Comprehensive NEET / JEE Engineering & Medical Core Revision
Module

Strictly Designed according to National Level Test Standards



IMPORTANT INSTRUCTION FOR STUDENTS (लाइले ध्यान दे): Ye koi normal summary sheet nahi hai. Isme Physics, Chemistry, aur Biology ke un sabhi 6 important core targets ko exhaustive depth me cover kiya gaya hai jo har saal competitive exams me repeat hote hain. Har table, trend order aur formula ko dhyani se padhein, yaad karein aur apni regular rough notebook me practice karein!

1. ELECTROSTATICS (स्थिरवैद्युतिकी) - FULL HIGH-YIELD COURSE

1.1 Electrostatic Charge & Properties

Charge matter ki fundamental property hai jiski wajah se forces develop hote hain. Charges do tarah ke hote hain: Positive (deficiency of electrons) aur Negative (excess of electrons).

Quantization Rule: Total charge hamesha integers me multiple hota hai basic discrete electron packet (e) ke:



Jahan $n \in \mathbb{N} (1, 2, 3...)$ aur $e = 1.602 \times 10^{-19} C$.

Conservation Principle: Kisi isolated system ka total net charge hamesha constant rehta hai, use generate ya destroy nahi kiya ja sakta, bas transfer ho sakta hai.

1.2 Coulomb's Law and Dielectric Effect

Do point charges ke beech lagne wala mechanical electrical force unke product ke directly proportional aur unke distance square ke inversely proportional hota hai.

Medium properties ke mutabik permittivity vary karti hai. Relative permittivity or Dielectric Constant:

Vacuum ke liye $k = 9 \times 10^9 N \cdot m^2 / C^2$. Water ke liye $K \approx 80$, Metals ke liye $K = \infty$.

1.3 Electric Field Intensity & Superposition Principle

Kisi specific continuous zone or spatial coordinates me electric field strength unit positive test charge par lagne wale force se balance hoti hai:

Agar multiple coordinates par discrete discrete charges loaded hain, to vector additions rule follow hote hain:

1.4 Electric Dipole Systems (Complete Equations)

Jab do matching scale scale magnitude ke charges opposite polarity me minimal micro-distance $2a$ standard scale par place hote hain to dipole form hota hai. Dipole Moment Vector:

Direction hamesha **Negative charge se Positive charge** ki taraf hoti hai.

Configuration / Position	Electric Field Vector Formula (E)	Electrostatic Potential (V)
Axial Line Position (End-on)	$E = 2kp / r^3$ (along p)	$V = kp / r^2$
Equatorial Line Position (Broadside)	$E = kp / r^3$ (opposite to p)	$V = 0$
General Point (At angle θ , distance r)	$E = (kp/r^3) \cdot \sqrt{(3\cos^2\theta + 1)}$	$V = kp \cdot \cos\theta / r^2$

1.5 Gauss's Theorem and Advanced Configurations

Total localized normal vector flux integration enclosed geometry boundaries ke scalar net value par dependent hoti hai:



Standard Application Formulations Grid:

- **Linear Charge Density (λ) Wire:** Cylindrical surface boundary symmetry checks give $E = \lambda / (2\pi\epsilon_0 r)$.
- **Surface Charge Density (σ) Infinite Non-conducting Sheet:** Distance independent value sheet $E = \sigma / (2\epsilon_0)$.
- **Conducting Shell Matrix:** Shell ke boundary regions coordinates checks: Outer region ($r > R$) $E = kQ/r^2$, Surface ($r = R$) $E = kQ/R^2$, Internal zone ($r < R$) $E = 0$.

2. MOLE CONCEPT & PERIODIC TABLE (रसायन विज्ञान कोर खंड)

2.1 Mole Concept Core Mathematics

Mole system parameters elements tracking ke liye absolute scale count values denote karta hai.

Avogadro constant limit value: $N_A = 6.022 \times 10^{23}$ particles per mol.

Triple Formula Matrix for Moles Calculation (n):

1. **By Weight Structure:** $n = \text{Weight in grams} / \text{Molar Mass (M)}$
2. **By Count Parameters:** $n = \text{Number of Entities} / 6.022 \times 10^{23}$
3. **By Ideal Gas Law Volume at STP parameters:** $n = \text{Volume of Gas (Liters)} / 22.4 \text{ L}$ (at standard 1 atm conditions) or 22.7 L (at 1 bar conditions).

2.2 Solution Concentration Engineering Terminology

Liquid and multi-solvent matrices solution quantitative calculations:

- **Molarity (M):** Temperature dependent metric scale. $M = \text{Moles of solute} / \text{Volume of solution in Liters}$.
- **Molality (m):** Temperature independent invariant system. $m = \text{Moles of solute} / \text{Mass of solvent in Kilograms}$.
- **Mole Fraction (X):** Absolute ratio parameter. $X_A = n_A / (n_A + n_B)$. Always remember $\sum X_i = 1$.

2.3 Modern Periodic Table & Periodic Laws

Moseley's Law state karta hai ki elements properties coordinate graphs unke nuclear structural active atomic number (Z) par dynamic reflect hoti hain:



Modern framework layout structural components are organized into 7 horizontal rows (periods) and 18 vertical columns (groups).

Critical Factor Trend Controls Across Periodic Table:

Property Element	Moving Left to Right Across a Period	Moving Top to Bottom Down a Group
Effective Nuclear Charge (Z_{eff})	Increases significantly due to addition of core electrons in same shell	Decreases or stays stable due to heavy shell screening effect
Atomic / Ionic Radii	Decreases (Shell content contracts tight)	Increases (New functional quantum shells add up)
Ionization Enthalpy (ΔH_{IE})	Increases (Requires extensive energy scales)	Decreases (Outer valence shell moves far away)
Electronegativity Scale (Paulings)	Increases (Maximum limits reached at Halogens)	Decreases down group tracking lines

3. BIOLOGY HIGH-YIELD ESSENTIAL PHYSIOLOGY & MECHANISMS

3.1 Body Fluids & Circulatory Core Systems

Human physiological body fluid balances system elements classification tracks:

- **Plasma Framework (55% share):** Yellow fluid containing 92% standard physiological water vector. Crucial Proteins balance tracks:
 - *Fibrinogen structural setups:* Inactive state cascade components for clotting mechanisms.
 - *Globulins:* Immunoglobulins structural base for specialized antibodies.
 - *Albumins scale:* Regulates crucial cellular water osmotic pressure balance matrix.
- **Formed Cellular Elements (45% share):**
 - *Erythrocytes (RBCs):* Biconcave, enucleated configuration. Count: 5 to 5.5 Million / mm^3 . Packed with Hemoglobin vector. Average operational life span metrics: 120 days. Scavenged and filtered out inside Spleen (Graveyard of RBC).
 - *Leukocytes (WBCs):* Immune systems response modules. Count: 6000-8000 / mm^3 . Classified into Neutrophils (highest percentage, phagocytic nature), Basophils (secrete Histamine, Serotonin, Heparin), Eosinophils (resist parasitic vector infections), and Lymphocytes (B & T cells for cell mediated / humoral immunity responses).
 - *Thrombocytes (Platelets):* Special megakaryocyte fragments tracking. Scale count: 1.5 to 3.5 Lakhs / mm^3 . Crucial blood clotting factor release points.

3.2 Excretory Products & Metabolic Elements Taxonomy

Nitrogenous dynamic metabolic side products degradation mapping inside living frameworks require standard pathways:

Absolute Biochemical Toxicity Spectrum Order:

Excretory Typology Mapping:

1. **Ammonotelic Matrix:** Requires massive amounts of aqueous vector streams to dissolve and clear out. Examples: Bony fishes, aquatic insects. Direct simple diffusion mechanisms via gill surfaces.
2. **Ureotelic Processing:** Ammonia is processed inside Liver channels via specialized cycle known as **Ornithine Cycle / Urea Cycle**. Energy consuming safe pathways. Examples: Mammals, terrestrial amphibians, marine chondrichthyes sharks.
3. **Uricotelic Adaptation Strategy:** Water conserving system optimization. Excreted in minimal water paste forms. Examples: Birds, land snails, desert reptiles.

3.3 Photosynthesis Essentials & Molecular Pathways

Anabolic bio-chemical transformation utilizing physical photons energy fluxes into hexose chemical carbohydrate stores:

Dual Phase Interdependent Kinetic Mechanisms:

- **1. Light Dependent Phase (Photochemical Engine):**

Occurs strictly inside specialized internal **Thylakoid Membrane matrix networks**. Involves multi-protein assembly photosystems (PS-II 680nm, PS-I 700nm). Photolysis of water releases protons and molecular oxygen gases:

Generates high chemical potential energy tracking vectors: **ATP** and **NADPH** via Z-scheme non-cyclic photophosphorylation channels.

- **2. Dark Biosynthetic Phase (Calvin Engine Carbon Tracking):**

Independent of direct lighting states but completely reliant on light phase stored compounds (**ATP / NADPH**). Occurs completely inside liquid **Stroma fields**.

Main Key Enzyme is **RuBisCO** (Most abundant protein globally). Core cycles track three critical physiological operational milestones:

1. *Carboxylation*: Fixation of raw atmospheric **CO₂** into stable 3-Phosphoglyceric Acid (3-PGA).
2. *Reduction*: Consumes 2 molecules of **ATP** and 2 molecules of **NADPH** per fixed **CO₂** to generate triose sugars.
3. *Regeneration*: Rebuilding of original acceptor molecule Ribulose 1,5-bisphosphate requiring additional 1 **ATP** metric scale.

Net Ledger Checklist for 1 Glucose Molecule (C₆H₁₂O₆): Calvin cycle loops exactly 6 times. Total biochemical fuel consumption requirements tally: **18 ATP molecules** and **12 NADPH molecules**.

4. MAHAMUKABLA INTENSE CLASSROOM WORKBOOK & DPP DRILLS

DPP SET-01: ELECTROSTATICS APPLICATION PATTERNS

Solve all problems by choosing the appropriate dimensional scalar or vector format. Detailed answers must match step-by-step with PCP methodology guides.

1. Two localized charges with magnitudes $+4\mu\text{C}$ and $-12\mu\text{C}$ are kept separated by distance d . If they are brought in physical contact and then placed back at same absolute separation distance coordinates, calculate the exact ratio of initial mechanical force to final operational mechanical force.
2. A thin spherical conducting matrix shell of radius R is loaded with a stable charge surface configuration Q . Find the exact work done in moving a micro test charge value q from the deep internal center coordinate location to a point on the peripheral surface boundary.
3. An electric dipole vector system consists of charges $\pm 2 \times 10^{-6} \text{ C}$ separated by stable baseline grid interval of 4mm . Calculate the magnitude of torque experienced when placed at alignment coordinate angles of 30° inside uniform field intensity grid of scale $5 \times 10^5 \text{ N/C}$.
4. Derive the electric field profile for an infinite sheet using Gauss's integration parameter bounding limits. Show graphically why variable distance coordinate parameters do not degrade or minimize field vectors.

DPP SET-02: MOLE CONCEPT & CALCULATIONS PRACTICE

1. Calculate the total net absolute count number of oxygen atoms present in exactly 44.8 Liters of pure dry ozone (O_3) gas tracked at standard STP conditions.
2. A solution matrix contains 40 grams of pure sodium hydroxide (NaOH) dissolved completely in 450 grams of pure distilled water solvent channels. Calculate both Molarity and Molality parameters given density values of resulting matrix is 1.2 g/mL .

3. Why does the ionization potential enthalpy scale of Nitrogen ($Z = 7$) surpass the ionization energy threshold scale values of Oxygen ($Z = 8$)? Explain thoroughly using half-filled stable p-orbital configuration arguments.

5.1 ADVANCED LEVEL SYLLABUS CORE DRILL - SESSION VARIANT 1

Comprehensive Review Grid (Deep Track Block 1)

This structural core analysis block provides dense technical tracking values for Revision Section #1. Practice sheets under PCP Sikar guidelines demand thorough data memorization schedules.

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Avogadro Quantum Index	$6.02214 \times 10^{23} \text{ ext{ particles/mol}}$	Invariant physical chemical structural scalar value.
RBC Physiological Pool Count	5.0 to 5.5 Million units per cubic millimeter	Deviations show clear anemial patterns.
Calvin Loop Metabolic Scale	18 molecules of ATP consumed per Hexose unit	Doubled requirement scales inside C_4 variant plant models.

PCP Smart Points Checklist:

- Electrostatics potential checks track fields along vector lines via derivatives: $E = -dV/dr$.
- Ornithine biochemical cycle transforms highly hazardous metabolic waste molecules into clean manageable streams via Carbon Dioxide fixation loops inside hepatocytes.
- RuBisCO enzyme showcases dual catalytic activity options: operational carboxylation alternates with hazardous photorespiratory oxygenation losses under heat stresses.

5.2 ADVANCED LEVEL SYLLABUS CORE DRILL - SESSION VARIANT 2

Comprehensive Review Grid (Deep Track Block 2)

This structural core analysis block provides dense technical tracking values for Revision Section #2. Practice sheets under PCP Sikar guidelines demand thorough data memorization schedules.

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5.3 ADVANCED LEVEL SYLLABUS CORE DRILL - SESSION VARIANT 3

Comprehensive Review Grid (Deep Track Block 3)

This structural core analysis block provides dense technical tracking values for Revision Section #3. Practice sheets under PCP Sikar guidelines demand thorough data memorization schedules.

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Permittivity Factor (ϵ) Tracker	$8.854 \times 10^{-12} \text{ ext{ C}^2/ \text{ext{N}\cdot\text{m}^2}}$	Scales lineally with medium constant multipliers K .
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5.4 ADVANCED LEVEL SYLLABUS CORE DRILL - SESSION VARIANT 4

Comprehensive Review Grid (Deep Track Block 4)

This structural core analysis block provides dense technical tracking values for Revision Section #4. Practice sheets under PCP Sikar guidelines demand thorough data memorization schedules.

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5.5 ADVANCED LEVEL SYLLABUS CORE DRILL - SESSION VARIANT 5

Comprehensive Review Grid (Deep Track Block 5)

This structural core analysis block provides dense technical tracking values for Revision Section #5. Practice sheets under PCP Sikar guidelines demand thorough data memorization schedules.

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5.6 ADVANCED LEVEL SYLLABUS CORE DRILL - SESSION VARIANT 6

Comprehensive Review Grid (Deep Track Block 6)

This structural core analysis block provides dense technical tracking values for Revision Section #6. Practice sheets under PCP Sikar guidelines demand thorough data memorization schedules.

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5.7 ADVANCED LEVEL SYLLABUS CORE DRILL - SESSION VARIANT 7

Comprehensive Review Grid (Deep Track Block 7)

This structural core analysis block provides dense technical tracking values for Revision Section #7. Practice sheets under PCP Sikar guidelines demand thorough data memorization schedules.

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5.8 ADVANCED LEVEL SYLLABUS CORE DRILL - SESSION VARIANT 8

Comprehensive Review Grid (Deep Track Block 8)

This structural core analysis block provides dense technical tracking values for Revision Section #8. Practice sheets under PCP Sikar guidelines demand thorough data memorization schedules.

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5.9 ADVANCED LEVEL SYLLABUS CORE DRILL - SESSION VARIANT 9

Comprehensive Review Grid (Deep Track Block 9)

This structural core analysis block provides dense technical tracking values for Revision Section #9. Practice sheets under PCP Sikar guidelines demand thorough data memorization schedules.

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5.10 ADVANCED LEVEL SYLLABUS CORE DRILL - SESSION VARIANT 10

Comprehensive Review Grid (Deep Track Block 10)

This structural core analysis block provides dense technical tracking values for Revision Section #10. Practice sheets under PCP Sikar guidelines demand thorough data memorization schedules.

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5.11 ADVANCED LEVEL SYLLABUS CORE DRILL - SESSION VARIANT 11

Comprehensive Review Grid (Deep Track Block 11)

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5.12 ADVANCED LEVEL SYLLABUS CORE DRILL - SESSION VARIANT 12

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5.13 ADVANCED LEVEL SYLLABUS CORE DRILL - SESSION VARIANT 13

Comprehensive Review Grid (Deep Track Block 13)

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5.14 ADVANCED LEVEL SYLLABUS CORE DRILL - SESSION VARIANT 14

Comprehensive Review Grid (Deep Track Block 14)

This structural core analysis block provides dense technical tracking values for Revision Section #14. Practice sheets under PCP Sikar guidelines demand thorough data memorization schedules.

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